

Skills Summary

Complete Root Sets and Correct Signs

Use this reference while completing your worksheet or when studying.

Skill 1 — Both square roots, every time

Rule: if $x^2 = k$ and $k > 0$, then $x = +\sqrt{k}$ and $x = -\sqrt{k}$.

Isolate x^2 first, then take the root of both sides and write the $\pm$ immediately — writing it later is when it gets forgotten.

Example: Solve $x^2 = 25$.

Step 1. The equation has no x term, so factoring is unnecessary — isolating x^2 and taking roots is the direct route, and it is the route where the negative root goes missing.

Step 2. $x = \pm\sqrt{25} = \pm 5$. Check both: $(-5)^2 = 25$ and $5^2 = 25$.

$$x = -5 \text{ or } x = 5$$

Try it: Solve $3x^2 = 27$.

$$x = 3 \text{ or } x = -3$$

(!) Watch out: $\sqrt{25}$ on its own means 5, the positive root. The second solution comes from the $\pm$ you write when you take the root of *both sides of an equation* — it is not part of the square-root symbol.

Skill 2 — The zero-product property

Rule: if $A \cdot B = 0$ then $A = 0$ or $B = 0$. Move everything to one side so the other side is 0, factor, then set each factor equal to 0.

Never divide both sides by x . That is division by something that may be zero, and it deletes the root $x = 0$.

Example: Solve $x^2 = 4x$.

Step 1. Both sides contain x , so the tempting move is to divide — instead move everything to one side, because the zero-product property keeps every root.

Step 2. $x^2 - 4x = 0$, so $x(x - 4) = 0$, giving $x = 0$ or $x - 4 = 0$.

$$x = 0 \text{ or } x = 4$$

Try it: Solve $x^2 + 3x = 0$.

$$x = 0 \text{ or } x = -3$$

Skill 3 — Signs in the quadratic formula

Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Write a , b , c *with their signs* before substituting, then write $-b$ on its own line. The denominator is $2a$, not 2: with $a = 3$ it is 6.

Example: Solve $x^2 - 5x + 6 = 0$.

Step 1. Three terms with a leading coefficient of 1 could be factored, but the formula never fails, so use it here and practise the signs.

Step 2. $a = 1$, $b = -5$, $c = 6$, so $-b = +5$ and $x = \frac{5 \pm \sqrt{25 - 24}}{2} = \frac{5 \pm 1}{2}$.
 $x = 2$ or $x = 3$

Try it: Solve $2x^2 + 3x - 2 = 0$.

check: $\frac{2}{1} = x$ or $\frac{3}{1} = x$